

## 21.4 Heteroscedasticity-consistent covariance estimators

Weighting needs a model for the variances. Often there is none worth trusting, and the analyst would rather keep ordinary least squares, which is unbiased whatever the variances are, and fix its standard errors. The covariance of  $\hat{\beta}$  under  $\text{Cov}(\varepsilon) = \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$  is

$$\text{Cov}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{i=1}^n \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1} \quad (21.4.1)$$

This is the covariance given by Theorem 19.2.2. The  $n$  variances cannot be estimated, one observation each. The idea of Eicker (1963), Huber (1967) and White (1980) is that they do not need to be. Only the  $p \times p$  matrix in the middle is needed, and a sum of  $n$  terms can be estimated consistently even when its terms cannot. The matrix is sandwiched between two copies of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , which gives the estimators their name.

### 21.4.1 The estimators

#### Definition 21.4.1. Heteroscedasticity-consistent covariance estimators

Let  $\hat{\varepsilon}_i$  and  $h_{ii}$  be the least squares residuals and leverages, with  $h_{ii} < 1$ . For weights  $\omega_i$ , put

$$\hat{\mathbf{V}}_\omega = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{i=1}^n \omega_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1}.$$

The estimators **HC0** to **HC3** take

$$\omega_i = \hat{\varepsilon}_i^2, \quad \frac{n}{n-p} \hat{\varepsilon}_i^2, \quad \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}, \quad \frac{\hat{\varepsilon}_i^2}{(1-h_{ii})^2}.$$

HC0 is White's estimator. HC1 applies the degrees-of-freedom correction of  $s^2$ . HC2 and HC3, proposed by MacKinnon and White (1985), correct each squared residual for its own leverage. Their finite-sample properties follow from Eq. 21.2.1 and from the deletion formulas of [Chapter 10](#).

#### Proposition 21.4.1. Finite-sample properties

Let  $\lambda \in \mathbb{R}^p$  and  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ , so that  $\lambda^\top \hat{\beta} - \lambda^\top \beta = \mathbf{c}^\top \varepsilon$  and  $\lambda^\top \hat{\mathbf{V}}_\omega \lambda = \sum_i c_i^2 \omega_i$ .

- (a) Under constant variance  $\sigma^2$ ,  $E(\lambda^\top \hat{\mathbf{V}}_{\text{HC2}} \lambda) = \text{Var}(\lambda^\top \hat{\beta})$ , while  $E(\lambda^\top \hat{\mathbf{V}}_{\text{HC0}} \lambda) = \text{Var}(\lambda^\top \hat{\beta}) - \sigma^2 \sum_i c_i^2 h_{ii} \leq \text{Var}(\lambda^\top \hat{\beta})$  and  $E(\lambda^\top \hat{\mathbf{V}}_{\text{HC3}} \lambda) \geq \text{Var}(\lambda^\top \hat{\beta})$ .
- (b) In general,  $E(\lambda^\top \hat{\mathbf{V}}_{\text{HC0}} \lambda) = \sum_i c_i^2 \sum_j (\mathbf{I} - \mathbf{M})_{ij}^2 \sigma_j^2$ .
- (c)  $\hat{\mathbf{V}}_{\text{HC0}} \preceq \hat{\mathbf{V}}_{\text{HC2}} \preceq \hat{\mathbf{V}}_{\text{HC3}}$  in the order of nonnegative definiteness.
- (d)  $\hat{\mathbf{V}}_{\text{HC3}} = \sum_{i=1}^n (\hat{\beta}_{(i)} - \hat{\beta})(\hat{\beta}_{(i)} - \hat{\beta})^\top$ , where  $\hat{\beta}_{(i)}$  is the estimate with observation  $i$  deleted.

*Proof.*

- (a) and (b).  $E\hat{\varepsilon}_i^2 = \sum_j (\mathbf{I} - \mathbf{M})_{ij}^2 \sigma_j^2$  by Eq. 21.2.1, which proves (b). Under constant variance it is  $\sigma^2(1 - h_{ii})$ , so the HC2 weights have expectation  $\sigma^2$  and  $E\sum_i c_i^2 \omega_i = \sigma^2 \|\mathbf{c}\|^2 = \text{Var}(\mathbf{c}^\top \boldsymbol{\varepsilon})$ . The HC0 weights have expectation  $\sigma^2(1 - h_{ii})$  and the HC3 weights  $\sigma^2/(1 - h_{ii}) \geq \sigma^2$ .
- (b) The HC weights satisfy  $\hat{\varepsilon}_i^2 \leq \hat{\varepsilon}_i^2/(1 - h_{ii}) \leq \hat{\varepsilon}_i^2/(1 - h_{ii})^2$  term by term, and each difference  $\hat{\mathbf{V}}_{\omega'} - \hat{\mathbf{V}}_{\omega}$  with  $\omega'_i \geq \omega_i$  is  $(\mathbf{X}^\top \mathbf{X})^{-1} \{\sum_i (\omega'_i - \omega_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top\} (\mathbf{X}^\top \mathbf{X})^{-1}$ , a nonnegative definite matrix.
- (c) By Proposition 10.6.2(a),  $\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ . The sum of the outer products of these vectors is  $\hat{\mathbf{V}}_{\text{HC3}}$ . ■

Part (d) says that HC3 measures how much the estimate moves when single observations are left out. This is the jackknife idea: the variability of an estimate is judged from its sensitivity to the data, not from a model for the errors. Part (a) makes HC0 biased downward when the model is right, and the bias is concentrated where  $c_i^2 h_{ii}$  is large, at high-leverage points that dominate the coefficient. Part (b) shows the same effect under heteroscedasticity: a high-leverage observation's squared residual understates its own variance, precisely when that variance matters most.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])
Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ food)
e = food - X @ beta
h = np.sum(Q**2, axis=1) # leverages
Rinv = np.linalg.inv(R) # (X'X)^{-1} = Rinv Rinv'

def sandwich(omega):
    """(X'X)^{-1} X' diag(omega) X (X'X)^{-1}, computed through Q."""
    meat = (Q * omega[:, None]).T @ Q
    return Rinv @ meat @ Rinv.T

p = X.shape[1]
covs = {
    "classical": (e @ e / (n - p)) * (Rinv @ Rinv.T),
    "HC0": sandwich(e**2),
    "HC1": sandwich(e**2) * n / (n - p),
    "HC2": sandwich(e**2 / (1 - h)),
    "HC3": sandwich(e**2 / (1 - h) ** 2),
}
for name, V in covs.items():
    print(f"{name:9s} slope se {np.sqrt(V[1, 1]):.4f}")
```

[Run this code online](#)

```
loo = np.array([np.linalg.lstsq(np.delete(X, i, 0), np.delete(food, i),
                             rcond=None)[0]
```

```

        for i in range(n)]])
# sum over i of (b_(i) - b)(b_(i) - b)'
jack = (loo - beta).T @ (loo - beta)
print("HC3 equals the sum of squared deletion changes:",
      np.allclose(jack, covs["HC3"]))
    
```

[Run this code online](#)

### Example. Standard errors for Engel's slope

For the least squares slope of food expenditure on income, the classical standard error  $s\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{22}^{1/2}$  is 0.0144. The heteroscedasticity-consistent standard errors are 0.0518 (HC0), 0.0520 (HC1), 0.0585 (HC2) and 0.0664 (HC3). The classical value is too small by a factor of 4.6 relative to HC3. Evidently the high-income households, which pin down the slope, are the noisy ones, so the classical formula, which averages the noise over all households, understates the uncertainty. The second listing above checks the jackknife identity of [Proposition 21.4.1\(d\)](#).

## 21.4.2 Consistency

The estimators are justified by a large-sample argument. We give it under simple conditions that make every step elementary: bounded regressors, errors with bounded fourth moments, and variances bounded away from zero. The theorem covers every  $\hat{\mathbf{V}}_\omega$  of [Definition 21.4.1](#).

### Theorem 21.4.2. Consistency of the sandwich estimators

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $p$  fixed. Assume that

- (H1) the errors are independent with mean zero and variances  $\sigma_i^2$ , with  $\sigma_i^2 \geq \sigma_{\min}^2 > 0$  and  $E\varepsilon_i^4 \leq K$  for all  $i$ ;
- (H2) the rows satisfy  $\|\mathbf{x}_{(i)}\| \leq C$  for all  $i$ ;
- (H3)  $n^{-1}\mathbf{X}^\top \mathbf{X} \rightarrow \mathbf{A}$ , a positive definite matrix.

Let  $\hat{\mathbf{V}}$  be any of HC0–HC3, and  $\mathbf{V} = \text{Cov}(\hat{\boldsymbol{\beta}})$  as in [Eq. 21.4.1](#). Then:

- $n(\hat{\mathbf{V}} - \mathbf{V}) \rightarrow \mathbf{0}$  in probability;
- for every fixed  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,  $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/(\boldsymbol{\lambda}^\top \hat{\mathbf{V}} \boldsymbol{\lambda})^{1/2} \rightarrow N(0, 1)$  in distribution;
- for every fixed  $p \times q$  matrix  $\boldsymbol{\Lambda}$  of rank  $q$ ,

$$W = (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\Lambda}^\top \boldsymbol{\beta})^\top (\boldsymbol{\Lambda}^\top \hat{\mathbf{V}} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\Lambda}^\top \boldsymbol{\beta})$$

tends to  $\chi^2(q)$  in distribution.

*Proof.* Write  $\mathbf{G}_n = n^{-1}\mathbf{X}^\top \mathbf{X}$ , and let  $\mathbf{B}_n = n^{-1} \sum_i \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  and  $\hat{\mathbf{B}}_n = n^{-1} \sum_i \omega_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  be the true and estimated middle matrices, so that  $n\mathbf{V} = \mathbf{G}_n^{-1} \mathbf{B}_n \mathbf{G}_n^{-1}$  and  $n\hat{\mathbf{V}} = \mathbf{G}_n^{-1} \hat{\mathbf{B}}_n \mathbf{G}_n^{-1}$ . Note  $\sigma_i^2 \leq (E\varepsilon_i^4)^{1/2} \leq K^{1/2}$ . By (H3),  $\mathbf{G}_n^{-1} \rightarrow \mathbf{A}^{-1}$ , and  $\lambda_{\min}(\mathbf{X}^\top \mathbf{X}) \geq cn$  for some  $c > 0$  and all large  $n$ .

*Step 1:  $\hat{\boldsymbol{\beta}}$  is consistent.*  $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr } \mathbf{V} \leq K^{1/2} \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} \leq K^{1/2} p / (cn) \rightarrow 0$ .

*Step 2: the errors estimate the middle.* Each entry of  $n^{-1} \sum_i (\varepsilon_i^2 - \sigma_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  is a sum of independent mean-zero terms with variance at most  $n^{-2} \sum_i K C^4 = K C^4 / n$ , so it tends to zero in probability by Chebyshev's inequality.

*Step 3: the residuals estimate the middle.* Let  $d_i = \mathbf{x}_{(i)}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , so  $\hat{\varepsilon}_i = \varepsilon_i - d_i$  and  $|d_i| \leq C \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|$ . Then, in the spectral norm,

$$\begin{aligned} \left\| n^{-1} \sum_i (\hat{\varepsilon}_i^2 - \varepsilon_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right\| &\leq \frac{C^2}{n} \sum_i (d_i^2 + 2|d_i||\varepsilon_i|) \\ &\leq C^4 \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 + 2C^3 \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| \frac{1}{n} \sum_i |\varepsilon_i|, \end{aligned}$$

which tends to zero because  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = o_p(1)$  and  $E n^{-1} \sum_i |\varepsilon_i| \leq K^{1/4}$ . With Step 2,  $\hat{\mathbf{B}}_n - \mathbf{B}_n \rightarrow \mathbf{0}$  for HC0.

*Step 4: the leverage corrections vanish.*  $h_{ii} \leq C^2 / \lambda_{\min}(\mathbf{X}^\top \mathbf{X}) \leq C^2 / (cn)$ , so  $\delta_n = (1 - \max_i h_{ii})^{-2} - 1 \rightarrow 0$ . For HC1–HC3,  $\hat{\varepsilon}_i^2 \leq \omega_i \leq (1 + \delta_n) \varepsilon_i^2$  for large  $n$  (with  $\delta_n$  replaced by  $p/(n-p)$  for HC1), so  $\mathbf{0} \preceq \hat{\mathbf{B}}_n - \hat{\mathbf{B}}_n^{\text{HC0}} \preceq \delta_n \hat{\mathbf{B}}_n^{\text{HC0}}$ , which tends to zero because  $\delta_n \rightarrow 0$  and  $\hat{\mathbf{B}}_n^{\text{HC0}} = \mathbf{B}_n + o_p(1)$  is bounded in probability, since  $\mathbf{0} \preceq \mathbf{B}_n \preceq K^{1/2} \mathbf{G}_n$  and  $\mathbf{G}_n$  converges. This proves (a), since  $n(\hat{\mathbf{V}} - \mathbf{V}) = \mathbf{G}_n^{-1}(\hat{\mathbf{B}}_n - \mathbf{B}_n)\mathbf{G}_n^{-1}$ .

(c) Let  $\mathbf{V}_\Lambda = \boldsymbol{\Lambda}^\top \mathbf{V} \boldsymbol{\Lambda}$  and  $\mathbf{T}_n = \mathbf{V}_\Lambda^{-1/2} \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , so that  $\text{Cov}(\mathbf{T}_n) = \mathbf{I}_q$ . For fixed  $\mathbf{a} \in \mathbb{R}^q$ ,  $\mathbf{a}^\top \mathbf{T}_n = \mathbf{c}^\top \boldsymbol{\varepsilon}$  with  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda} \mathbf{V}_\Lambda^{-1/2} \mathbf{a}$ . By the Cauchy–Schwarz inequality in the metric  $(\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $c_i^2 \leq h_{ii} \|\mathbf{c}\|^2$ , so  $\max_i c_i^2 / \|\mathbf{c}\|^2 \leq \max_i h_{ii} \rightarrow 0$ . Lemma 21.2.1(b), with  $\delta = 2$ , shows that  $\mathbf{a}^\top \mathbf{T}_n / \|\mathbf{a}\| \rightarrow N(0, 1)$  for  $\mathbf{a} \neq \mathbf{0}$ . By the Cramér–Wold device,  $\mathbf{T}_n \rightarrow N_q(\mathbf{0}, \mathbf{I}_q)$ . Next,  $n\mathbf{V}_\Lambda = \boldsymbol{\Lambda}^\top \mathbf{G}_n^{-1} \mathbf{B}_n \mathbf{G}_n^{-1} \boldsymbol{\Lambda} \succeq \sigma_{\min}^2 \boldsymbol{\Lambda}^\top \mathbf{G}_n^{-1} \boldsymbol{\Lambda}$ , whose limit  $\sigma_{\min}^2 \boldsymbol{\Lambda}^\top \mathbf{A}^{-1} \boldsymbol{\Lambda}$  is positive definite; and  $n\mathbf{V}_\Lambda$  is bounded because  $\mathbf{B}_n \preceq K^{1/2} \mathbf{G}_n$ . So  $(n\mathbf{V}_\Lambda)^{-1/2}$  is bounded, and by (a)

$$\begin{aligned} \mathbf{V}_\Lambda^{-1/2} (\boldsymbol{\Lambda}^\top \hat{\mathbf{V}} \boldsymbol{\Lambda}) \mathbf{V}_\Lambda^{-1/2} &= \mathbf{I}_q + (n\mathbf{V}_\Lambda)^{-1/2} \boldsymbol{\Lambda}^\top \\ &\quad \times n(\hat{\mathbf{V}} - \mathbf{V}) \boldsymbol{\Lambda} (n\mathbf{V}_\Lambda)^{-1/2} \rightarrow \mathbf{I}_q. \end{aligned}$$

Since  $W = \mathbf{T}_n^\top \{ \mathbf{V}_\Lambda^{-1/2} (\boldsymbol{\Lambda}^\top \hat{\mathbf{V}} \boldsymbol{\Lambda}) \mathbf{V}_\Lambda^{-1/2} \}^{-1} \mathbf{T}_n$ , continuous mapping and Slutsky's lemma give  $W \rightarrow \mathbf{T}^\top \mathbf{T}$  with  $\mathbf{T} \sim N_q(\mathbf{0}, \mathbf{I}_q)$ , which is  $\chi^2(q)$ . Part (b) is the case  $q = 1$ , with the sign kept. ■

The theorem does not assume that the errors are normal, that their variances follow any model, or that they are identically distributed. It does assume independence, which the next two sections relax. The bounded regressors of (H2) are stronger than needed. Eicker (1967) and White (1980) replace them by conditions that amount to  $\max_i h_{ii} \rightarrow 0$  together with moment bounds, and White's version allows random regressors drawn independently from a common distribution, which covers most observational data. We do not prove these extensions.

In practice the statistic in (b) is compared with the  $t(n-p)$  distribution rather than the normal. That choice has no exact justification, but it costs nothing asymptotically and helps in small samples. A heteroscedasticity-robust test of  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  compares  $W/q$  with  $F(q, n-p)$ .

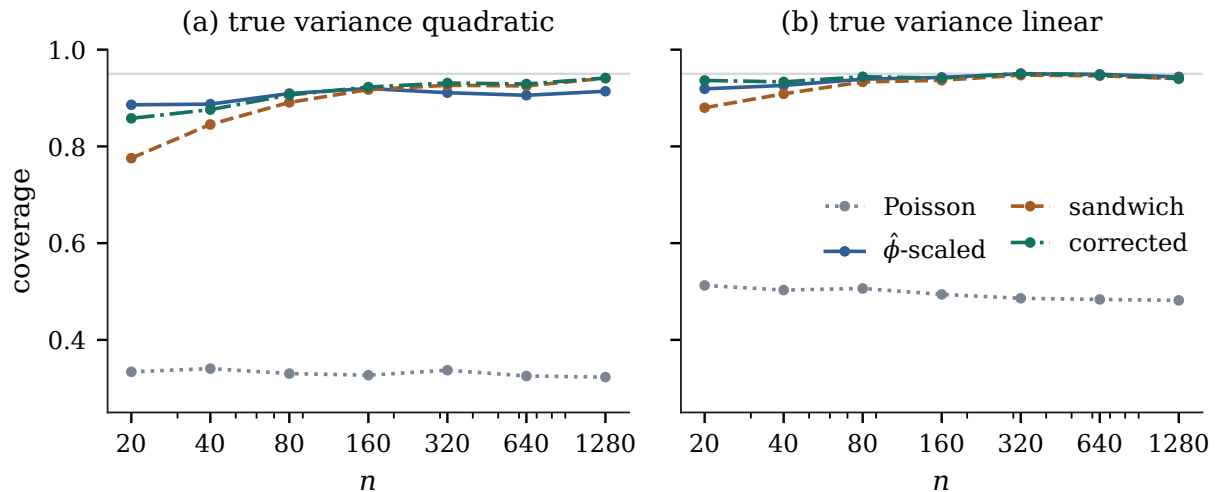


Figure 21.4.1: Coverage of nominal 95% intervals for the slope in a design with lognormal quantiles as regressor values. (a) Constant error variance. (b) Error standard deviation proportional to  $x$ . HC1 is omitted; it lies just above HC0.

### 21.4.3 Finite-sample behaviour

Consistency says nothing about how large  $n$  must be. The answer depends on the leverages, as [Proposition 21.4.1](#) suggests, and it can be sobering.

#### Example. Coverage in a skewed design

Let the regressor values be the  $n$  quantiles  $x_i = \exp\{\Phi^{-1}((i - 1/2)/n)\}$  of a standard lognormal distribution, a design with a long right tail like incomes. The largest leverage is 0.58 for  $n = 25$  and still 0.15 for  $n = 1000$ . [Figure 21.4.1](#) shows the coverage of nominal 95% intervals for the slope, with normal errors, over 20,000 simulated data sets for each  $n$ .

With constant variance (panel a) the classical interval is exact, and the robust intervals pay a price for their robustness: at  $n = 25$  they cover 0.852 (HC0), 0.866 (HC1), 0.895 (HC2) and 0.932 (HC3). With standard deviation proportional to  $x$  (panel b) the classical interval fails badly. Its coverage is 0.456 at  $n = 25$  and falls to 0.287 at  $n = 1000$ , because the growing right tail of the design gives ever more influence to the noisiest observations. The robust intervals improve with  $n$ , but slowly. At  $n = 25$  they cover 0.631 (HC0), 0.650 (HC1), 0.760 (HC2) and 0.853 (HC3); at  $n = 1000$ , 0.912, 0.912, 0.924 and 0.935. HC3 is the best of the four throughout, as Long and Ervin (2000) found in a wide range of designs.

```
rng = np.random.default_rng(2107)

def coverage(n, hetero, reps):
    """Coverage of nominal 95% slope intervals, classical and HC0-HC3."""
    # lognormal quantiles
    x = np.exp(stats.norm.ppf((np.arange(1, n + 1) - 0.5) / n))
    Xs = np.column_stack([np.ones(n), x])
    Qs, Rs = np.linalg.qr(Xs)
    hs = np.sum(Qs**2, axis=1)
    # slope estimate - slope = c @ errors
    c = Qs @ np.linalg.inv(Rs).T[:, 1]
    E = rng.normal(size=(reps, n)) * (x if hetero else 1.0)
```

```

err = E @ c
res = E - (E @ Qs) @ Qs.T
var = {"classical": (res**2).sum(axis=1) / (n - 2) * (c @ c),
      "HC0": (res**2) @ c**2,
      "HC1": (res**2) @ c**2 * n / (n - 2),
      "HC2": (res**2 / (1 - hs)) @ c**2,
      "HC3": (res**2 / (1 - hs) ** 2) @ c**2}
q = stats.t.ppf(0.975, n - 2)
cover = {k: np.mean(np.abs(err) <= q * np.sqrt(v)) for k, v in var.items()}
return cover, hs.max()

for n_sim in (25, 100, 400):
    cov, hmax = coverage(n_sim, hetero=True, reps=2000)
    print(f"n = {n_sim:3d} (max leverage {hmax:.2f}):",
          {k: round(v, 3) for k, v in cov.items()})

```

[Run this code online](#)

## 21.4.4 Leverage and the sandwich

The poor small-sample behaviour has a simple source. The variance estimate  $\sum_i c_i^2 \omega_i$  is a weighted sum of squared residuals, and when a few weights  $c_i^2$  dominate, it behaves like a variance estimate with very few degrees of freedom, however large  $n$  is. A  $t(n-p)$  reference distribution then has tails that are far too light.

### Example. One household carries the standard error

In Engel's data the household with the largest income has leverage 0.255. Its term  $c_i^2 \hat{\varepsilon}_i^2 / (1 - h_{ii})^2$  makes up 0.85 of the HC3 variance of the slope. The robust standard error of [Example \(Standard errors for Engel's slope\)](#) is, in effect, an estimate of one household's error variance. That is not a reason to go back to the classical standard error, which is far worse. It is a reason to look at that household (the influence measures of [Chapter 20](#), such as Definition 20.4.1, are the tools), to report how the conclusions change without it, and to prefer weighting, which here gives that household much less influence (Example (Engel's data, reweighted)).

Several refinements address the degrees-of-freedom problem. Bell and McCaffrey's (2002) adjustment, as recommended by Imbens and Kolesár (2016), combines HC2 with a Satterthwaite approximation to the degrees of freedom of  $\sum_i c_i^2 \omega_i$ . The wild bootstrap of [Chapter 23](#) (Definition 23.2.1) resamples the residuals with random signs, keeping each one attached to its own row, and often gives better calibrated intervals than any of the HC formulas.

## 21.4.5 Exercises

### 21.4.5.1 A. Check your understanding

#### Exercise. A1

Take the two-group model  $y_{gj} = \mu_g + \varepsilon_{gj}$ ,  $g = 1, 2$ ,  $j = 1, \dots, n_g$ . Show that HC2 for  $\hat{\mu}_1 - \hat{\mu}_2$  equals  $s_1^2/n_1 + s_2^2/n_2$ , the variance estimate of Welch's test, where  $s_g^2$  is the sample variance of group  $g$ . What does HC0 give?

*Solution.* Here  $h_{ii} = 1/n_g$ ,  $\hat{\varepsilon}_{gj} = y_{gj} - \bar{y}_g$ , and  $\mathbf{c}$  has entries  $1/n_1$  in group 1 and  $-1/n_2$  in group 2. HC2 is  $\sum_g \sum_j \hat{\varepsilon}_{gj}^2 / \{(1 - 1/n_g)n_g\} = \sum_g \sum_j (y_{gj} - \bar{y}_g)^2 / \{(n_g - 1)n_g\} = \sum_g s_g^2/n_g$ . HC0 gives  $\sum_g (n_g - 1)s_g^2/n_g^2$ , which is smaller.

### Exercise. A2

What happens to HC0, HC2 and HC3 if some observation has  $h_{ii} = 1$ ? Explain why no estimator built from residuals can estimate that observation's contribution to  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ .

## 21.4.5.2 B. Practice

### Exercise. B1

In simple regression with an intercept and constant variance, show that the relative bias of HC0 for the slope variance is  $-\sum_i (x_i - \bar{x})^2 h_{ii} / S_{xx}$ . Evaluate it for  $x = (1, 2, 3, 4, 10)$  of Exercise (A1).

*Solution.* For the slope,  $c_i = (x_i - \bar{x})/S_{xx}$  and  $\|\mathbf{c}\|^2 = 1/S_{xx}$ , so Proposition 21.4.1(a) gives relative bias  $-\sum_i c_i^2 h_{ii} / \|\mathbf{c}\|^2 = -\sum_i (x_i - \bar{x})^2 h_{ii} / S_{xx}$ . With  $S_{xx} = 50$  and the leverages  $(0.38, 0.28, 0.22, 0.20, 0.92)$ , the sum is  $9(0.38) + 4(0.28) + 1(0.22) + 0 + 36(0.92) = 37.88$ , and the relative bias is  $-37.88/50 = -0.76$ . HC0 estimates the slope variance at a quarter of its true value, on average, because the point  $x = 10$  dominates the slope and has a tiny residual.

### Exercise. B2

Using Theorem 21.3.1(b), write down a sandwich estimator of the covariance of a weighted least squares estimate with fixed weights  $\mathbf{W}$ , and show that it reduces to HC0 applied to the rescaled data  $(\mathbf{W}^{1/2}\mathbf{X}, \mathbf{W}^{1/2}\mathbf{y})$ .

### Exercise. B3

White's test compares  $s^2 \mathbf{X}^\top \mathbf{X}$  with  $\sum_i \hat{\varepsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ . Show that the difference of these matrices, divided by  $n$ , has entries  $n^{-1} \sum_i (\hat{\varepsilon}_i^2 - s^2) x_{ij} x_{ik}$ , and explain why Koenker's statistic with  $\mathbf{z}_i$  the distinct products  $x_{ij} x_{ik}$  tests whether these are jointly zero.

## 21.4.5.3 C. Going deeper

### Exercise. C1

Suppose the rows  $(\mathbf{x}_{(i)}, \varepsilon_i)$  are independent copies of a random pair with  $E(\varepsilon | \mathbf{x}) = 0$ ,  $E\|\mathbf{x}\|^4 < \infty$ ,  $E\varepsilon^4 < \infty$  and  $E\mathbf{x}\mathbf{x}^\top$  positive definite. Adapt the proof of Theorem 21.4.2 to show that HC0 is consistent in the sense of part (a), replacing Chebyshev's inequality by the weak law of large numbers.

**Exercise. C2**

The usual jackknife variance is  $\frac{n-1}{n} \sum_i (\hat{\boldsymbol{\beta}}_{(i)} - \mathbf{m})(\hat{\boldsymbol{\beta}}_{(i)} - \mathbf{m})^\top$ , where  $\mathbf{m}$  is the mean of the  $\hat{\boldsymbol{\beta}}_{(i)}$ . Express it in terms of  $\hat{\mathbf{V}}_{\text{HC3}}$  and the vector  $\sum_i \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ , and show that the two agree to first order when  $\max_i h_{ii} \rightarrow 0$ .